



Startups

“To Grow To THE NEXT LEVEL, We Must Build CRITICAL IPs”



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India's semiconductor ambitions hinge on fostering domestic demand for indigenous designs. But who will pave the way? The success of fabless giants like Qualcomm and Nvidia offers a blueprint. EFY's Yashasvini Razdan spoke to Sridhar Kaip of Semiconductor Fabless Accelerator Lab (SFAL) on what it takes to reach that level.

With your experience working with fabless startups, how do you evaluate India's current position in the global semiconductor ecosystem?

A useful way to understand India's position is through a pyramid model. At the top are mature markets, such as the US, Europe, and Japan, supporting approximately 2000 companies in the fabless semiconductor space. Below them are ecosystems such as Israel and Taiwan, with around 1000 companies. India, however, currently has fewer than 150 startups in this domain.

Over the past five to six years, India's semiconductor ecosystem has shown promising progress, but it is still in its early stages. Karnataka, especially Bengaluru, holds a unique advantage due to its concentration of captive centres, multinational companies, and abundant semiconductor talent. However, in terms of homegrown products and intellectual property (IP), we remain at an early stage.

What steps are needed to reach the next level of growth in this domain?

To achieve the next level, we must build critical IPs. This is a key focus area for SFAL. Some initiatives will involve leveraging existing resources, while others will require strategic development and expansion of core competencies.

The government of India has already announced initiatives to support this effort, and several startups are stepping forward to contribute. Belgium's journey 40 years ago provides valuable inspiration. Despite having no significant presence in electronics or semiconductors, Belgium initiated government-backed programmes to support the ecosystem. Today, it generates approximately \$2 billion in revenue from a self-sustaining semiconductor industry.

What challenges do fabless startups in India face?

Most startups face common

challenges in the pre-idea and early stages. However, for fabless startups, the primary challenge is investment. Most investors seek quick returns, but the gestation period for ROI in the semiconductor industry is more extended. This makes funding a significant challenge, as even the tape-out of a simple test chip costs between ₹5 million and ₹10 million.

This expenditure often comes after 18 months of development—for a simple industrial application chip, not a complex one.

What role does the government play in removing these barriers and facilitating growth?

The Ministry of Electronics and Information Technology (MeitY) and the Karnataka government have engaged in early-stage discussions and incorporated learnings from initiatives like SFAL into the Design Linked Incentive (DLI) and Chips to Startup (C2S) programmes. These efforts are pivotal in